

Problem Set #2

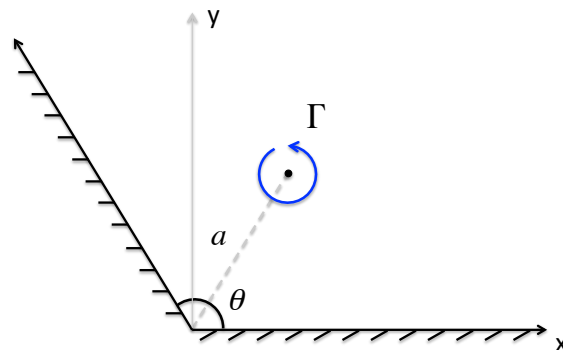
Due Fri., Mar. 20 by 11:59pm

Please hand in your completed problem sets as PDF files to the course Canvas site. Working together with others is fine (and encouraged), but you must write up each solution yourself.

1. A common exam question in undergraduate fluid mechanics is to find a purely real velocity potential function given a streamfunction for a steady, incompressible, irrotational, inviscid flow. While complex potential functions are not discussed, undergraduate students can still solve this type of problem by finding the velocity components and then integrating.

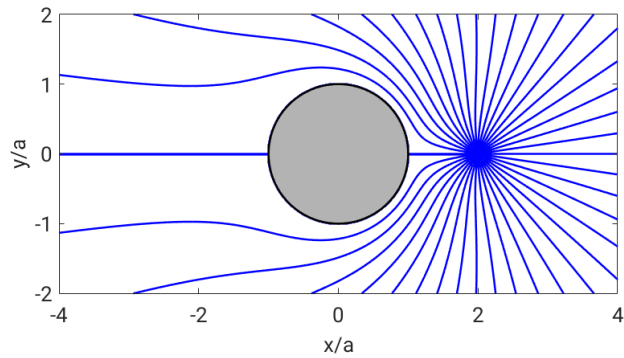
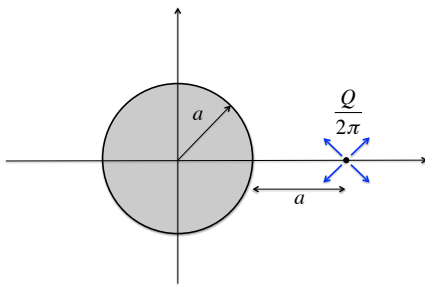
- Find a complex potential function for a flow having exactly four stagnation points at the locations $(x, y) = (1, 0)$, $(0, 1)$, $(-1, 0)$, and $(0, -1)$, which are the “4th roots of unity.”
- From this complex potential function, express the real velocity potential $\phi(x, y)$ and streamfunction $\psi(x, y)$ as a purely real functions of x and y .
- Using your favorite software (e.g., matlab), plot streamlines and potential lines for this flow, and verify that the flow has stagnation points at the above three points.
- Using this info, write a question appropriate for an undergraduate fluid mechanics exam.

2. An irrotational vortex is placed in a corner formed by two walls meeting at an angle of $\theta = 120^\circ$. The vortex is equidistant from the two walls and located a distance a away from the corner. The fluid circulation around a small contour enclosing the vortex center is Γ . Assume that the resulting flow is two-dimensional.



- Find a complex potential function for inviscid flow in the corner.
- Find an expression for velocity along the positive x -axis. Verify that the positive x -axis corresponds to a streamline.
- Using your favorite software, plot potential lines and streamlines of the flow inside the corner, assuming $a = 1$ and $\Gamma = 1$.

3. Consider a source of strength $\frac{Q}{2\pi}$ placed at a distance of $2a$ away from the center of cylinder of radius a .



- Find a complex potential function describing this flow.
- Evaluate the force (magnitude and direction) that the source exerts on the cylinder. What happens to this force if the source is replaced by a sink?
- Use your favorite software to recreate the streamline plot shown in the right-hand figure above.
- Extra credit:** Adapt the BEM solver that you developed in problem set 1 to compute the same streamlines as in part (c) and compare them on your plot to the analytic solution using the complex potential function you have derived in part (a).